MTD-Playground: An Attacker-Aware Evaluation Framework for Network Moving Target Defense

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Abstract

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CCS Concepts

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Keywords

Moving Target Defense, Software Defined Networking, Game Theory, AI/ML, Computer & Network Security

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1 INTRODUCTION

Modern networks face increasingly sophisticated cyber threats, including reconnaissance, poisoning, and distributed denial-of-service (DDoS) attacks, which exploit static network configurations and predictable host identifiers [21][2][29]. Rather than exploiting a single vulnerability, attackers typically follow multi-stage attack campaigns that begin with reconnaissance and initial compromise of externally exposed services, followed by credential theft, lateral movement across internal systems, and ultimately the exfiltration of sensitive data. These attack strategies exploit the static nature of traditional network infrastructures, where system configurations, network paths, and service locations remain largely unchanged over time. Once an attacker learns the structure of the network during reconnaissance, this knowledge can be reused to plan and execute subsequent attack stages with relatively low uncertainty. Traditional defense mechanisms typically rely on static configurations, which allow attackers sufficient time to analyze vulnerabilities and launch targeted attacks. Rules and signature based mechanisms such as firewalls, IDS/IPS, and redundancy, are often inadequate against persistent and adaptive adversaries, particularly in distributed or cyber-physical systems [3][13]. To address these challenges, Moving Target Defense (MTD) has emerged as a proactive approach that dynamically transforms the attack surface to increase attacker uncertainty and reduce the probability of successful exploitation [20][40]. The core idea of MTD is to continuously change aspects of the system such as IP addresses, network paths, or service configurations in order to invalidate attacker knowledge and increase the difficulty of performing successful reconnaissance and lateral movement. As a result, attackers must repeatedly restart reconnaissance and exploitation processes, increasing their operational cost and reducing the probability of successful compromise.

Software-Defined Networking (SDN) provides a natural platform for network-level MTD due to centralized control, global visibility, and programmability. SDN enables efficient implementation of diverse MTD strategies, including IP mutation or randomization, path randomization, and deception-based defenses. IP-based MTD techniques periodically assign virtual IP addresses to hosts or services to obscure network identities and prevent reconnaissance or scanning attacks [21][20]. Path randomization modifies forwarding paths dynamically to prevent attackers from predicting network reachability [18][1], while deception-based approaches generate virtual network views, deploy decoy hosts, or redirect traffic to mislead adversaries during reconnaissance [2]. Recent works explore hybrid and distributed MTD architectures to improve scalability and performance. Multi-controller SDN frameworks enable frequent IP shuffling without overloading a single controller [35], while hybrid solutions combine address and path mutation or leverage SmartNICs for low-overhead deployment [18]. Domain-specific applications of SDN-based MTD have been demonstrated in critical infrastructures such as smart grids and DNS networks, improving resilience against DoS and flooding attacks while maintaining service continuity [1][8]. For cyber-physical systems, adaptive MTD schemes employ reinforcement learning to balance security, resource usage, and system performance [13].

Beyond SDN, game-theoretic methods have been applied to model attacker–defender strategy interactions for Moving Target Defense, using empirical game analysis to evaluate defensive configurations and strategic behavior in dynamic network environments [36]. Stackelberg and other game-theoretic frameworks have been used to derive optimal timing and configuration strategies for MTD, analyzing trade-offs between defensive cost and attack success probability in strategic settings [26]. These models provide strategic insights for minimizing attack success probability while accounting for defense costs. Meanwhile, AI/ML-driven MTD leverages

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data-driven models to detect emerging attack patterns, predict optimal mutation schedules, and adaptively select defensive actions in real time. Recent work demonstrates that attackers can use unsupervised learning to infer and fingerprint MTD trigger times and mutation patterns, motivating the need for designs that reduce information leakage [33]. Deep reinforcement learning has been used to adaptively select MTD policies based on observed network states and attacker responses, providing context-aware defense strategies that balance security effectiveness with resource constraints [16]. Federated multi-agent deep reinforcement learning has been explored to coordinate MTD across distributed environments, enabling adaptive defense actions against dynamic adversaries [44]. AI/ML techniques can complement SDN-based mechanisms by dynamically tuning IP, path, and deception parameters based on observed network behavior and attacker activity.

Despite significant research progress in MTD mechanisms, evaluating their effectiveness remains a major challenge. Existing studies often focus on individual techniques, such as IP hopping, path randomization, or topology mutation, and evaluate them using custom experimental environments or simplified network models. These evaluation setups typically differ in network topology, attacker assumptions, experimental metrics, and threat scenarios, making it difficult to directly compare results across different studies. Furthermore, many prior evaluations focus on isolated attack steps rather than realistic multi-stage attack campaigns that reflect how adversaries operate in modern enterprise environments. This lack of standardized evaluation environments creates a significant barrier to understanding the practical effectiveness of MTD mechanisms. Without a controlled and repeatable evaluation framework, it is difficult for researchers and practitioners to determine which MTD strategies are most effective under realistic attack conditions. In particular, there is currently no widely adopted platform that enables systematic comparison of different network MTD approaches such as SDN-based randomization, AI/ML-driven adaptive defenses, or game-theoretic strategies under consistent enterprise style network deployments and multi-stage attack scenarios.

To address this gap, we present **MTD-Playground**, an attacker-aware evaluation framework designed to benchmark network Moving Target Defense (MTD) mechanisms in a controlled and repeatable environment. The framework models a realistic enterprise network architecture consisting of an external network, perimeter firewall, public-facing DMZ web server, internal application server, database server, and internal client workstations. Within this environment, attackers perform multi-stage intrusion campaigns that include reconnaissance, initial compromise, lateral movement, privilege escalation, and data exfiltration. MTD-Playground provides standardized interfaces that allow different MTD mechanisms to be deployed over the same logical topology and evaluated under identical attack conditions. This design enables fair, reproducible comparisons across diverse MTD approaches while capturing realistic attacker–defender dynamics. By enabling systematic evaluation of MTD mechanisms under consistent attack scenarios, MTD-Playground helps researchers and practitioners better understand the strengths, limitations, and operational trade-offs of different dynamic defense strategies. The framework also provides a foundation for future research on adaptive and attacker-aware defense mechanisms that can effectively disrupt multi-stage attacks in enterprise networks.

Contributions. The main contributions of this work are summarized as follows:

- We identify the lack of standardized benchmarking environments as a key limitation in existing MTD research and highlight the challenges of comparing different MTD mechanisms across inconsistent experimental setups.
- We design and implement *MTD-Playground*, an attacker-aware evaluation framework that enables systematic benchmarking of network Moving Target Defense mechanisms under realistic enterprise-style network deployments.
- We model multi-stage attack campaigns that capture realistic adversarial behavior, including reconnaissance, initial compromise, lateral movement, and data exfiltration.
- We provide a controlled experimental platform that supports repeatable experiments and fair comparison of diverse MTD strategies using consistent metrics and attack scenarios.

2 BACKGROUND & MOTIVATION

This section provides background on Moving Target Defense (MTD) and Software-Defined Networking (SDN). We first introduce the fundamental concepts of MTD and its role. We then discuss SDN, as a practical platform for implementing dynamic defense mechanisms such as path randomization used in this work. Later we provide our motivation section here.

MTD: Moving Target Defense (MTD) is a proactive security paradigm that aims to reduce the effectiveness of cyber attacks by continuously changing aspects of a system's attack surface. Traditional systems often maintain static configurations, allowing attackers sufficient time to perform reconnaissance, develop exploits, and plan multi-stage intrusions. In contrast, MTD introduces controlled and dynamic changes to system attributes, thereby increasing uncertainty for attackers and limiting their ability to obtain reliable knowledge of the target environment [23][45]. MTD strategies are commonly characterized along three design dimensions: *what* to change, *how* to change it, and *when* the changes should occur. The *what* dimension refers to the system components that can be dynamically modified, such as IP addresses, network paths, ports, services, or virtual machines. The *how* dimension describes the mechanisms used to perform these changes, including address mutation, path randomization, service migration, and deception-based techniques. Finally, the *when* dimension determines the triggering policy for reconfiguration, which may be periodic, event-driven, or adaptive based on system state or observed attacker behavior [12][39]. By continuously altering system configurations, MTD techniques aim to increase attack cost, disrupt reconnaissance activities, and reduce the probability of successful exploitation in complex network environments [45][39].

SDN: Software-Defined Networking (SDN) is a network architecture that separates the control plane from the data plane, enabling centralized and programmable network management. In SDN environments, the **control plane** determines how traffic should be handled by computing routing decisions and network policies, whereas the **data plane** consists of forwarding devices such as switches that process and forward packets based on rules installed by the

controller. Communication between the controller and switches is typically implemented using protocols such as OpenFlow [31]. The centralized control and programmability provided by SDN make it a suitable platform for implementing dynamic security mechanisms. Network policies, routing paths, and access control rules can be updated programmatically by the controller without requiring manual reconfiguration of individual network devices. This flexibility allows rapid adaptation of network behavior in response to evolving threats or operational requirements [2][35][20]. In the context of Moving Target Defense, SDN enables efficient implementation of dynamic reconfiguration strategies such as IP address mutation, path randomization, and traffic redirection. By dynamically updating forwarding rules in network switches, the SDN controller can modify communication paths and network visibility, making it more difficult for attackers to perform reliable reconnaissance or maintain persistent access within the network [21].

Some of the papers from introduction will be added here.

Evaluation Challenges and the Need for Evaluation Platforms. Despite significant progress in Moving Target Defense (MTD) research, a key challenge remains the lack of standardized evaluation environments that enable fair and repeatable comparison across different approaches. Existing studies [**cite**] often evaluate their mechanisms within isolated experimental setups using different network topologies, attacker assumptions, and evaluation metrics. For example, SDN-based MTD techniques are frequently tested using simplified network environments or custom infrastructures, while many game-theoretic approaches rely primarily on analytical modeling or simulations rather than deployment in realistic networks. Similarly, AI/ML-driven MTD strategies are commonly evaluated using simulated environments or synthetic datasets that may not fully capture the complexity of real-world network traffic and attack behavior. Consequently, results across these studies are difficult to compare, and it remains unclear how different MTD strategies perform relative to one another under consistent attack scenarios. This limitation motivates the need for a common experimental platform that enables systematic evaluation and benchmarking of diverse MTD mechanisms under consistent network conditions and attack scenarios.

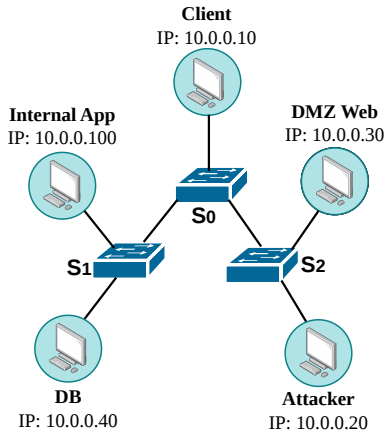

Figure 1: Topology of our Enterprise Network System.

3 RELATED WORK

Prior work can generally be categorized into three main research directions: Software-Defined Networking (SDN)-based MTD, Artificial Intelligence and Machine Learning (AI/ML)-driven MTD, and Game-theoretic MTD strategies which have proposed in this literature.

SDN-based MTD. Software-Defined Networking provides a natural platform for implementing network-level MTD due to its centralized control and programmability. IP mutation, shuffling and address randomization techniques has been studied, where hosts are periodically assigned virtual IP addresses to obscure real network identities and disrupt reconnaissance or scanning attacks [21][29][20]. To improve scalability and performance, introduced distributed or hybrid mutation architectures, including multi-controller SDN environments and hybrid mutation strategies combining address and random path changes [18][35]. Similarly, IPv6-based connection shuffling has been explored to leverage the larger address space available in modern networks [30]. Deception-based defenses were investigated, where SDN controllers generate virtual network views or decoy hosts to mislead attackers during reconnaissance [2]. Additionally, SDN-based MTD mechanisms have also been applied in domain specific environments, such as smart grid communication networks and DNS infrastructures, where dynamic routing or service migration helps mitigate denial-of-service attacks [1][8]. These mechanisms are typically implemented through SDN controllers that periodically update forwarding rules and network reachability relationships.

Game-Theoretic-based MTD. Game-theoretic models provide a formal framework for analyzing strategic interactions between attackers and defenders in dynamic MTD systems [41]. Early work evaluates MTD mechanisms using equilibrium-based analysis to quantify attack success probability and defense effectiveness under rational attacker assumptions [11, 37]. Stackelberg game formulations have been proposed to model defender commitment and attacker best-response behavior, particularly for optimizing mutation timing and reconfiguration intervals [7]. Deception-oriented Stackelberg models further analyze strategic interaction under attacker observability constraints [28]. To capture multi-stage attack progression, several studies adopt stochastic and Markov game frameworks, where system states evolve based on joint attacker–defender actions [15, 42]. Robust Markov game formulations additionally incorporate uncertainty in attacker behavior and system transitions [6]. Game-theoretic MTD has also been applied to domain-specific infrastructures, including critical embedded systems [9] and cyber-physical power grid environments [24]. More recently, reinforcement learning has been integrated with Markov game formulations to enable adaptive policy learning under dynamic conditions [4, 5]. Despite these advancements, most game-theoretic MTD approaches remain evaluated in simulation-based environments

and are not tightly integrated with programmable SDN enforcement platforms.

AI/ML-based MTD. Recent work has shifted MTD from static periodic shuffling to learning-guided, context-aware adaptation. Early SDN-MTD studies demonstrated that address mutation and controller-orchestrated reconfiguration can reduce reconnaissance and follow-on exploitation, but they largely relied on fixed/random schedules and did not optimize timing under dynamic adversaries [19, 22, 34]. AI/ML-based methods then introduced policy learning over network state and attacker signals. DQ-MOTAG uses deep Q-learning to adapt service-migration timing under DDoS pressure, explicitly balancing security gain and reconfiguration cost [14]. Adversarial deep RL extends this direction by training robust policies against strategic or adaptive opponents [17]. At larger scale, federated multi-agent DRL enables distributed coordination while preserving locality of data and control, making adaptive MTD more practical for multi-domain enterprise settings [43]. In parallel, hybrid strategic-learning lines combine game-theoretic attacker-defender modeling with adaptive execution, improving policy realism compared with purely static equilibria [27, 38]. Importantly, MTDSense shows that low-entropy or predictable MTD can be fingerprinted with high accuracy, highlighting that adaptation quality (not just mutation frequency) is central to real security gains [32]. In contrast, our work is not another new MTD mechanism; we build a systematic evaluation environment where AI/ML and non-learning MTD approaches are tested under identical enterprise-style conditions, attacker models, and metrics, enabling controlled, repeatable, and fair comparison of effectiveness, overhead, and robustness across prior techniques.

4 MTD-PLAYGROUND FRAMEWORK DESIGN

4.1 System & Threat Model

System Model. We evaluate MTD mechanisms in a compact but realistic enterprise-style network designed to mimic the essential structure of real-world multi-stage intrusions in which the attacker must progress through successive phases of reconnaissance, initial compromise, credential harvesting or stuffing, lateral movement, and data ex-filtration to achieve their objective while enabling controlled and repeatable experimentation. The system follows a topology consisting of an external network, a perimeter firewall, a public-facing DMZ web server, an internal application server, a database server containing sensitive data, and a small number of internal client workstations. This topology reflects common enterprise deployment patterns and naturally supports multi-stage intrusions, which are central to modern attack campaigns. We focus on representative network-level MTD techniques, specifically SDN-based path and flow randomization, as this mechanisms are widely studied and deployed and capture the core design space of network MTD. In the SDN-based setting, a centralized controller dynamically modifies forwarding paths and flow rules to create time-varying network views and reachability relationships. The evaluation platform provides standardized interfaces and metrics that allow different commercial, open-source, and research MTD implementations to be deployed over the same logical topology and subjected to identical attack campaigns, enabling fair, repeatable, and comparison of similar network structure that is largely missing from prior MTD evaluations.

Threat Model. We consider a motivated, external, multi-stage attacker consistent with contemporary enterprise threat models. The attacker originates outside the enterprise network and has standard offensive capabilities, including reconnaissance, initial compromise (exploitation), lateral movement, and so on. The attacker's primary objective is to compromise the database server and successfully ex-filtrate sensitive data. Attack success is defined by unauthorized access to the database server, execution of unauthorized database queries, and successful ex-filtration of a predefined dataset. Intermediate milestones include compromise of the DMZ web server, pivoting to the internal application server, and establishment of lateral movement paths. We focus on full attack campaigns rather than isolated single-step exploits, as MTD is intended to disrupt attacker progress across multiple phases of intrusion rather than prevent a single vulnerability from being exploited. Unlike many prior MTD studies that assume static or naive attackers, we explicitly model a partial-knowledge, MTD-aware adversary. The attacker is assumed to be aware that the target follows an enterprise architecture and may employ MTD, but does not know the internal topology, current IP addresses, randomized network state, or reconfiguration schedule, and has no access to defender-side control-plane state or telemetry. The attacker is adaptive and may repeat reconnaissance and adjust strategy based on observed failures, but is not assumed to fully reverse-engineer the MTD mechanism or compromise the defender's control plane. The defender controls network configuration, firewall and routing policies, and MTD reconfiguration logic, and has access to standard operational telemetry including network flow records, firewall logs, host logs on managed servers, and SDN controller telemetry when applicable. We do not assume perfect detection, and the evaluation is designed to isolate the impact of MTD dynamics on attacker progress and cost. Overall, this threat model balances realism and experimental control, allowing us to rigorously measure how MTD affects attacker progress and cost across full multi-stage attack campaigns under conditions that closely mirror practical enterprise deployments.

We consider a multi-tier architecture consisting of an Internet-facing web server, an internal application server, and a backend database. The attacker's objective is to compromise deeper tiers and ultimately exfiltrate sensitive data from the database. To model realistic multi-stage/hop adversarial behavior, we treat Remote Code Execution (RCE), Server-Side Request Forgery (SSRF), and credential-based administrative compromise as reusable attack primitives rather than single-stage attacks. These primitives may be applied across multiple transitions in the attack chain, such as Internet-to-Web, Web-to-App, and App-to-DB, depending on the attacker's capabilities and the reachable attack surface.

In addition to multi-stage compromise attacks, we incorporate DDoS as an availability-oriented threat targeting the web tier. Unlike RCE, SSRF, and credential-based attacks, which aim to progressively compromise internal services, DDoS operates as a parallel attack that overwhelms system resources. In realistic adversarial settings, DDoS may also serve as a distraction mechanism, masking

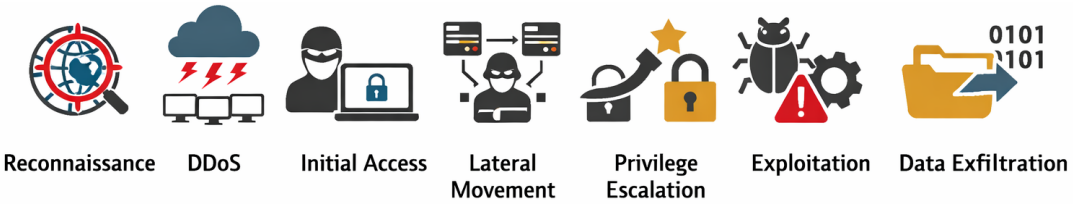

Figure 2: Multi-stage attack workflow assumed in this work to evaluate MTD effectiveness, illustrating adversarial progression from reconnaissance to data exfiltration.

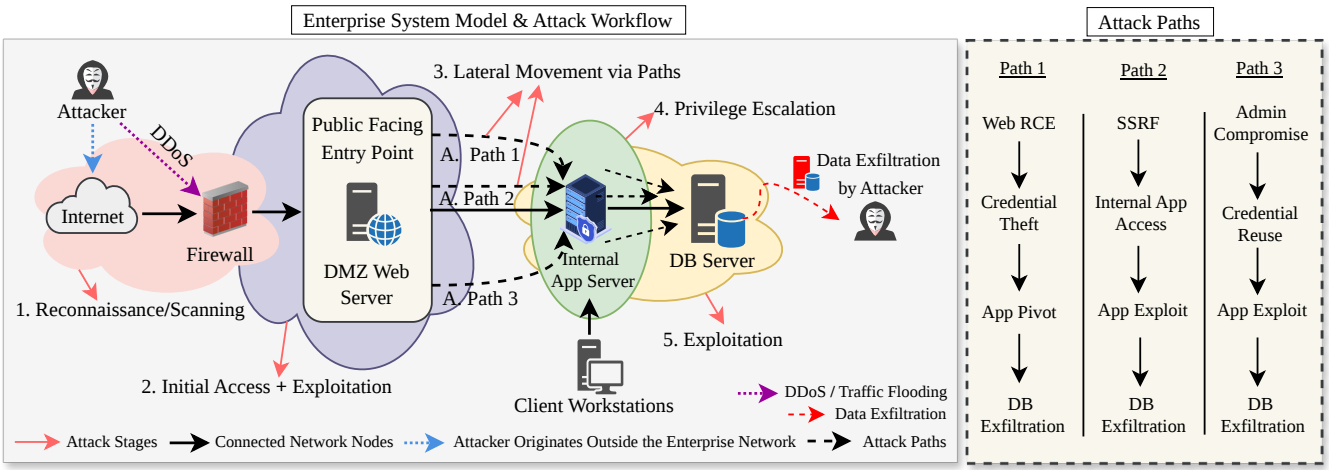

Figure 3: Enterprise System Model and Multi-Stage Attack Workflow.

Table 1: Combined view of attack types, behaviors, and MTD effects.

Attack Primitives	Goal	Behavior	MTD Effect
RCE / SSRF / Admin Comp.	Data theft	Stealthy, multi-hop	Disrupt pivoting
DDoS	Service disruption	High-volume flooding	Improve resilience

stealthy multi-hop attacks. This allows us to evaluate the effectiveness of MTD mechanisms under both confidentiality and availability threats.

We model a multi-stage attack path over a multi-tier architecture (Internet → Web → Application → Database), where an adversary progresses through stages including reconnaissance, initial access, lateral movement, and privilege escalation before ultimately exfiltrating sensitive data. These stages are not strictly sequential and may occur in different combinations depending on the attacker's capabilities and available attack surface.

Possible Path 1: Exploit-driven path
Recon → Initial Access on Web (RCE) → Lateral Movement to App → Privilege Escalation or Credential Theft → DB Access → Data Exfiltration

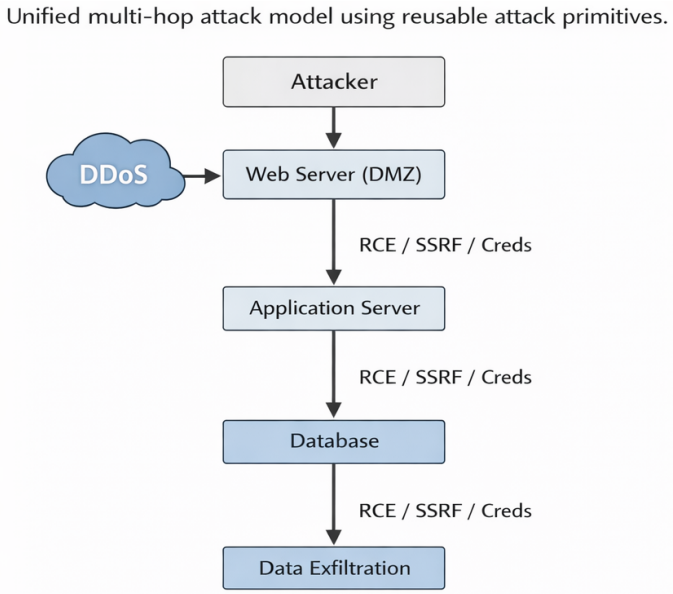

Figure 4: Unified multi-hop attack model.

Explanation: The attacker first performs reconnaissance to identify vulnerabilities in the public web application. Using an exploit

(e.g., RCE), they gain code execution on the web server and establish a foothold. From this compromised host, the attacker performs lateral movement to reach the internal application server. To access sensitive resources, they escalate privileges or extract credentials from the system. Finally, the attacker accesses the database and exfiltrates sensitive data.

Possible Path 2: SSRF-driven path
Recon → Initial Access on Web → SSRF-based Reachability to App → App Compromise or Credential Abuse → DB Access → Data Exfiltration

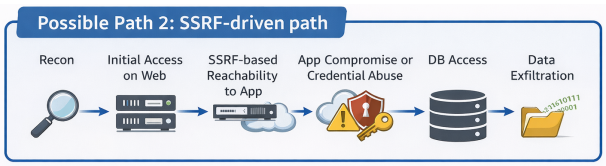

Figure 5: Possible attack path 2 (As an Example).

Explanation: The attacker identifies a web feature that can be abused to trigger server-side requests (SSRF). Without fully compromising the web server, they leverage it to access internal services such as the application server. This bypasses network restrictions by abusing trusted communication paths. The attacker then exploits the application or abuses exposed credentials/tokens. Once access is established, they reach the database and extract sensitive data.

Possible Path 3: Identity-driven path
Recon → Credential Compromise → Privileged Web Access → Lateral Movement to App → Privilege Escalation if needed → DB Access → Data Exfiltration

Explanation: The attacker obtains valid credentials through methods such as phishing, leaks, or credential reuse during reconnaissance. These credentials provide privileged access to the web server (e.g., admin panel or system access). From this trusted position, the attacker performs lateral movement to the internal application server. If required, they escalate privileges or reuse additional credentials to reach the database. Finally, they access and exfiltrate sensitive data.

5 MTD MECHANISM UNDER EVALUATION

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6 EXPERIMENTAL SETUP

To evaluate the effectiveness of the proposed Moving Target Defense (MTD) framework, we implemented a prototype testbed using a software-defined networking (SDN) emulation environment. The experimental environment is built on a virtualized network platform that integrates network emulation, container-based services, and centralized SDN control.

6.1 Emulation Environment

The experiments were conducted using Mininet[1] [25] running in a virtual machine environment to emulate the enterprise network topology. To support containerized services and realistic application deployment, Containernet[2] was used as an extension of Mininet. Containernet enables Docker[3] containers to operate as network hosts while maintaining compatibility with Mininet's network emulation features. The SDN infrastructure uses Open vSwitch[4] (OVS) as the data plane switch, while the control plane is managed by the ONOS[5] [10] SDN controller. The controller is responsible for installing forwarding rules, managing network flows, and dynamically enforcing MTD actions such as path mutation and traffic redirection.

6.2 Network System Topology

The experimental network represents a simplified enterprise architecture consisting of five primary hosts implemented as Docker containers within Containernet: **(1) Attacker node** – simulates an external adversary performing multi-stage (e.g., recon, lateral movement) attack attempts. **(2) Client node** – represents legitimate user traffic accessing enterprise services. **(3) DMZ Web Server** – provides publicly accessible web services and acts as the first entry point for external traffic. **(4) Application Server** – processes application logic and communicates with backend services. **(5) Database Server** – stores enterprise data and represents the most critical internal asset and it is the attacker's ultimate goal to reach and exfiltrate the data. Importantly, The attacker, web server, application server, and database server are deployed as Docker-based containers using Containernet, allowing realistic service deployment and isolation while maintaining lightweight resource usage. The client is implemented as a standard Mininet host to emulate a normal user endpoint interacting with the enterprise services. These hosts are interconnected through Open vSwitch switches, forming a multi-layer enterprise network architecture that includes a demilitarized zone (DMZ) and an internal network segment.

Here we have used three switches for simplification but it can be customizable based on topology anyone want to create. A single controller is used to simplify the control-plane design and ensure deterministic experimentation while still enabling dynamic network reconfiguration through SDN. The selection of five hosts is intended to capture a representative multi-tier enterprise architecture commonly used in web-based services. Such architectures typically involve externally accessible services in a DMZ, internal application processing layers, and protected database systems. This setup also enables the emulation of multi-stage attack scenarios—including reconnaissance, initial compromise, and lateral movement which are frequently considered in MTD research. By modeling these components, the experimental environment provides a realistic yet controlled platform for evaluating the effectiveness of different MTD strategies.

[1] Mininet: http://mininet.org/
[2] Containernet: https://containernet.github.io/
[3] Docker: https://www.docker.com/
[4] Open vSwitch: https://www.openvswitch.org/
[5] ONOS: https://onosproject.org/

6.3 Security Components

Layer-2 Firewall. An ONOS Layer-2 (L2) firewall is deployed to enforce network access control policies by filtering Ethernet frames based on parameters such as MAC addresses, switch ports, and traffic direction. This mechanism restricts unauthorized communication between hosts and prevents direct access from external entities to internal network segments unless explicitly permitted, reflecting common enterprise security practices. The firewall is managed through the SDN controller, enabling dynamic updates of filtering rules during experimentation. Integrating the L2 firewall within the SDN-controlled network allows the testbed to model realistic security enforcement while supporting the evaluation of Moving Target Defense (MTD) mechanisms under dynamic network reconfiguration.

6.4 Attack Simulation

To rigorously evaluate the resilience of SDN-based path randomization, attack scenarios are simulated from a dedicated attacker host within the experimental testbed. These scenarios replicate realistic adversarial behavior in enterprise networks, targeting key components such as the DMZ web server, application server, and database server. The attack simulation framework is designed to assess both preventive and adaptive defense effectiveness of the Moving Target Defense (MTD) mechanism.

Attack Workflow. The attacker follows a multi-stage workflow, reflecting common intrusion patterns observed in real-world enterprise attacks:

(1) **Network Recon/Scanning:** The attacker performs host discovery, port scanning, and service enumeration to map the network topology. Path randomization aims to obfuscate the network structure, making reconnaissance results inconsistent and unreliable. The attacker may also attempt to launch a DDoS attack targeting the firewall in order to overwhelm its filtering capacity and bypass security controls. If successful, this may allow the attacker to disrupt normal network operations and increase the likelihood of reaching the DMZ web server.
(2) **Initial Access:** Once potential targets (Public Facing Entry Point) are identified, the attacker attempts to exploit web application services running on the DMZ web server. This evaluates how dynamic path changes and firewall policies affect an attacker's ability to reach services.
(3) **Lateral Movement:** Following compromise of DMZ web server, the attacker attempts to move laterally toward internal resources using those three attack paths, particularly the internal application server. The randomized network paths disrupt predictable routes, increasing the complexity and effort required for successful lateral movement.
(4) **Privilege Escalation:** Following a successful service compromise, the attacker attempts to escalate privileges on the host to gain higher-level access. This simulates the transition from limited user access to administrative control, enabling further attacks on internal confidential resources.
(5) **Service Exploitation:** After identifying accessible services, the attacker attempts to exploit vulnerabilities in the database server, which represents the final target in the attack chain. This stage simulates attempts to gain unauthorized access to sensitive data stored in the backend system.

6.5 Path Randomization MTD Deployment

The Moving Target Defense (MTD) mechanism is deployed within the SDN controller, where path mutation policies are executed based on predefined strategies. The controller dynamically updates forwarding rules in Open vSwitch switches to alter the communication paths between hosts. Instead of maintaining static routing paths, the controller periodically randomizes the network routes used for communication among enterprise components such as the client, web server, application server, and database server. Path randomization has been widely explored as an effective MTD technique because it reduces network predictability and increases the uncertainty faced by attackers during reconnaissance and lateral movement [**cite**]. Leveraging the programmability of Software Defined Networking (SDN) enables efficient implementation of such dynamic path mutation without requiring modifications to end hosts. As a result, the proposed framework uses SDN-based path randomization as the underlying mechanism for evaluating different MTD strategies within a controlled experimental environment.

7 EVALUATION

7.1 Evaluation Metrics

To systematically evaluate the effectiveness of Moving Target Defense (MTD) mechanisms within the proposed MTD-Playground, we define a set of metrics that capture both *security effectiveness* and *system performance*. These metrics are designed to enable fair comparison across different MTD strategies under consistent experimental conditions.

7.2 Quantitative Metrics

1) Attack Success Rate (ASR). Attack Success Rate measures the proportion of attack attempts that successfully achieve their intended objective, such as reaching the internal application server or exfiltrating data from the database. A lower ASR indicates stronger defense effectiveness.

$$ASR = \frac{\text{Number of Successful Attacks}}{\text{Total Attack Attempts}}$$

2) Attack Completion Time (ACT). Attack Completion Time represents the time required for an attacker to complete a multi-stage attack (e.g., reconnaissance to data exfiltration). Effective MTD mechanisms increase ACT by disrupting attacker progress.

3) Reconnaissance Accuracy. This metric measures how accurately an attacker can discover valid hosts, services, or paths in the network during scanning. It is computed as the ratio of correctly identified targets to the total number of actual targets. Lower accuracy indicates better obfuscation.

4) Path Stability / Path Mutation Rate. Path mutation rate quantifies how frequently network paths change due to MTD policies. Higher mutation rates increase unpredictability but may introduce overhead. This metric helps analyze the trade-off between security and stability.

5) Network Unpredictability (Entropy). Entropy measures the randomness introduced by MTD in network configurations (e.g.,

path selection). Higher entropy indicates greater uncertainty for attackers.

$$H = -\sum p_i \log p_i$$

where p_i is the probability of selecting a specific path or configuration.

6) End-to-End Latency. Latency measures the delay experienced by legitimate traffic between communicating hosts. It is typically computed using round-trip time (RTT). MTD mechanisms should minimize latency overhead while maintaining security.

7) Packet Loss Rate. This metric captures the percentage of packets lost during transmission. Frequent path changes or misconfigurations may increase packet loss, negatively affecting service quality.

8) Throughput. Throughput measures the amount of successfully delivered data over the network per unit time. It evaluates whether MTD mechanisms degrade network performance under normal or attack conditions.

9) Controller Overhead. Controller overhead measures the computational and communication load imposed on the SDN controller due to MTD operations, including flow rule updates and path reconfigurations. It can be quantified in terms of CPU utilization, memory usage, and number of flow modifications.

10) Flow Rule Update Rate. This metric measures how frequently forwarding rules are updated in switches. High update rates indicate dynamic reconfiguration but may impact scalability and controller performance.

11) Service Availability. Service availability measures whether legitimate users can successfully access services (e.g., web, application, database) during MTD operations and attack scenarios. It is typically expressed as the percentage of successful client requests.

12) Detection and Mitigation Time. This measures the time required to detect abnormal behavior (if applicable) and apply MTD actions (e.g., path mutation). Faster response improves defense effectiveness.

7.3 Qualitative Metrics

1) Defense Effectiveness under Multi-stage Attacks. Evaluates how well MTD disrupts different attack stages such as reconnaissance, lateral movement, and data exfiltration.

2) Trade-off Analysis (Security vs. Performance). Analyzes the balance between improved security and system overhead.

3) Policy Conflict Analysis. Examines conflicts between security policies (e.g., firewall rules vs. path changes) that may impact correctness or security.

4) Attack Graph / Attack Tree Analysis. Models possible attack paths and evaluates how MTD disrupts attacker progression through the system.

5) Risk Metrics. Assesses system risk based on vulnerability exposure, attack likelihood, and potential impact.

6) CIA Metrics. Evaluates the impact of MTD on Confidentiality, Integrity, and Availability of the system.

7) Cost Metrics. Evaluates deployment and operational cost of MTD mechanisms, including infrastructure and management overhead.

8) Quality of Service (QoS). Assesses user experience aspects such as stability, reliability, and service continuity under MTD operations.

Overall, these metrics enable a comprehensive evaluation of MTD mechanisms in terms of their ability to disrupt attackers while maintaining acceptable network performance and scalability.

Table 2: Evaluation Metrics Summary

Metric	Category
Attack Success Rate	Security
Attack Completion Time	Security
Reconnaissance Accuracy	Security
Entropy	Security
Latency	Performance
Throughput	Performance
Controller Overhead	System
Flow Update Rate	System
Availability	Reliability

8 DISCUSSION

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9 LIMITATIONS & FUTURE WORK

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10 CONCLUSION

References

[1] Moataz Abdelkhalek, Burhan Hyder, Manimaran Govindarasu, and Craig G Rieger. 2022. Moving Target Defense Routing for SDN-enabled Smart Grid. In *2022 IEEE International Conference on Cyber Security and Resilience (CSR)*. 215–220. doi:10.1109/CSR54599.2022.9850341

[2] Stefan Achleitner, Thomas La Porta, Patrick McDaniel, Shridatt Sugrim, Srikanth V. Krishnamurthy, and Ritu Chadha. 2016. Cyber Deception: Virtual Networks to Defend Insider Reconnaissance. In *Proceedings of the 8th ACM CCS International Workshop on Managing Insider Security Threats* (Vienna, Austria) *(MIST '16)*. Association for Computing Machinery, New York, NY, USA, 57–68. doi:10.1145/2995959.2995962

[3] Noor O. Ahmed and Bharat Bhargava. 2016. Mayflies: A Moving Target Defense Framework for Distributed Systems. In *Proceedings of the 2016 ACM Workshop on Moving Target Defense* (Vienna, Austria) *(MTD '16)*. Association for Computing Machinery, New York, NY, USA, 59–64. doi:10.1145/2995272.2995283

[4] Author1 and Author2. Year. Multi-Agent Reinforcement Learning for Moving Target Defense Decision-Making Based on Markov Game. *Journal/Conference Name* (Year), Start–End. doi:DOI

[5] Author1 and Author2. Year. Optimal Strategy for Moving Target Defense on the Internet of Vehicles Based on Game Theory and Reinforcement Learning. *Journal/Conference Name* (Year), Start–End. doi:DOI

[6] Author1 and Author2. Year. Optimal Strategy Selection Approach to Moving Target Defense Based on Markov Robust Game. *Journal/Conference Name* (Year), Start–End. doi:DOI

[7] Author1 and Author2. Year. Optimal Timing of Moving Target Defense: A Stackelberg Game Model. *Journal/Conference Name* (Year), Start–End. doi:DOI

[8] Abdullah Aydeger, Pei Zhou, Sanzida Hoque, Marco Carvalho, and Engin Zeydan. 2025. MTDNS: Moving Target Defense for Resilient DNS Infrastructure. In *2025 IEEE 22nd Consumer Communications Networking Conference (CCNC)*. 1–6. doi:10.1109/CCNC54725.2025.10975971

[9] Maxime Ayrault, Étienne Borde, Ulrich Kühne, and Jean Leneutre. 2021. Moving Target Defense Strategy in Critical Embedded Systems: A Game-theoretic Approach. In *2021 IEEE 26th Pacific Rim International Symposium on Dependable Computing (PRDC)*. 27–36. doi:10.1109/PRDC53464.2021.00014

[10] Pankaj Berde, Marco Gerola, Jonathan Hart, Yuta Higuchi, Masayoshi Kobayashi, Toshiyuki Koide, Bob Lantz, Brian O'Connor, Pavel Radoslavov, William Snow, et al. 2014. ONOS: Towards an open, distributed SDN OS. In *Proceedings of the*

Third Workshop on Hot Topics in Software Defined Networking (HotSDN). ACM, 1–6.

[11] Gui-lin Cai, Bao-sheng Wang, and Qian-qian Xing. 2017. Game Theoretic Analysis for the Mechanism of Moving Target Defense. *Frontiers of Information Technology Electronic Engineering* 18, 12 (2017), 2017–2034. doi:10.1631/FITEE.1601797

[12] Marco Carvalho and Richard Ford. 2014. Moving-target defenses for computer networks. *IEEE Security & Privacy* 12, 2 (2014), 73–76.

[13] Xinzhong Chai, Yasen Wang, Chuanxu Yan, Yuan Zhao, Wenlong Chen, and Xiaolei Wang. 2020. DQ-MOTAG: Deep Reinforcement Learning-based Moving Target Defense Against DDoS Attacks. In *2020 IEEE Fifth International Conference on Data Science in Cyberspace (DSC)*. 375–379. doi:10.1109/DSC50466.2020.00065

[14] Xinzhong Chai, Yasen Wang, Chuanxu Yan, Yuan Zhao, Wenlong Chen, and Xiaolei Wang. 2020. DQ-MOTAG: Deep Reinforcement Learning-Based Moving Target Defense Against DDoS Attacks. In *2020 IEEE Conference on Dependable and Secure Computing (DSC)*. doi:10.1109/DSC50466.2020.00065

[15] Ankur Chowdhary, Sailik Sengupta, Adel Alshamrani, Dijiang Huang, and Abdulhakim Sabur. 2019. Adaptive MTD Security using Markov Game Modeling. In *2019 International Conference on Computing, Networking and Communications (ICNC)*. 577–581. doi:10.1109/ICCNC.2019.8685647

[16] Taha Eghtesad, Yevgeniy Vorobeychik, and Aron Laszka. 2020. Adversarial Deep Reinforcement Learning Based Adaptive Moving Target Defense. In *Decision and Game Theory for Security: 11th International Conference, GameSec 2020, College Park, MD, USA, October 28–30, 2020, Proceedings* (College Park, MD, USA). Springer-Verlag, Berlin, Heidelberg, 58–79. doi:10.1007/978-3-030-64793-3_4

[17] Taha Eghtesad, Yevgeniy Vorobeychik, and Aron Laszka. 2020. Adversarial Deep Reinforcement Learning Based Adaptive Moving Target Defense. In *Decision and Game Theory for Security (GameSec)*. 258–279. doi:10.1007/978-3-030-64793-3_4

[18] Mateusz Fronczyk and Mariusz Rawski. 2025. Hybrid mutation MTD solution with dedicated SDN agents. In *2025 20th Conference on Computer Science and Intelligence Systems (FedCSIS)*. 145–154. doi:10.15439/2025F3136

[19] C. Gudla and A. Sung. 2020. Moving Target Defense Discrete Host Address Mutation and Analysis in SDN. In *2020 International Conference on Computational Science and Computational Intelligence (CSCI)*. doi:10.1109/CSCI51800.2020.00017

[20] Charan Gudla and Andrew H. Sung. 2020. Moving Target Defense Discrete Host Address Mutation and Analysis in SDN. In *2020 International Conference on Computational Science and Computational Intelligence (CSCI)*. 55–61. doi:10.1109/CSCI51800.2020.00017

[21] Jafar Haadi Jafarian, Ehab Al-Shaer, and Qi Duan. 2015. An Effective Address Mutation Approach for Disrupting Reconnaissance Attacks. *IEEE Transactions on Information Forensics and Security* 10, 12 (2015), 2562–2577. doi:10.1109/TIFS.2015.2467358

[22] J. H. Jafarian, E. Al-Shaer, and Q. Duan. 2015. An Effective Address Mutation Approach for Disrupting Reconnaissance Attacks. *IEEE Transactions on Information Forensics and Security* 10, 12 (2015), 2562–2577. doi:10.1109/TIFS.2015.2467358

[23] Sushil Jajodia, Anup K Ghosh, Vipin Swarup, Cliff Wang, and X Sean Wang. 2011. *Moving target defense: creating asymmetric uncertainty for cyber threats*. Springer Science & Business Media.

[24] Subhash Lakshminarayana, E. Veronica Belmega, and H. Vincent Poor. 2021. Moving-Target Defense Against Cyber-Physical Attacks in Power Grids via Game Theory. *IEEE Transactions on Smart Grid* 12, 6 (2021), 5244–5257. doi:10.1109/TSG.2021.3095083

[25] Bob Lantz, Brandon Heller, and Nick McKeown. 2010. A network in a laptop: Rapid prototyping for software-defined networks. In *Proceedings of the 9th ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets)*. ACM, 1–6.

[26] Henger Li, Wen Shen, and Zizhan Zheng. 2020. Spatial-Temporal Moving Target Defense: A Markov Stackelberg Game Model. In *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems* (Auckland, New Zealand) *(AAMAS '20)*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 717–725.

[27] Henger Li, Wen Shen, and Zizhan Zheng. 2020. Spatial-Temporal Moving Target Defense: A Markov Stackelberg Game Model. In *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*.

[28] Duohe Ma, Zhimin Tang, Xiaoyan Sun, Lu Guo, Liming Wang, and Kai Chen. 2022. Game Theory Approaches for Evaluating the Deception-based Moving Target Defense. In *Proceedings of the 9th ACM Workshop on Moving Target Defense* (Los Angeles, CA, USA) *(MTD'22)*. Association for Computing Machinery, New York, NY, USA, 67–77. doi:10.1145/3560828.3563995

[29] Saumil Macwan and Chung-Horng Lung. 2019. Investigation of Moving Target Defense Technique to Prevent Poisoning Attacks in SDN. In *2019 IEEE World Congress on Services (SERVICES)*, Vol. 2642-939X. 178–183. doi:10.1109/SERVICES.2019.00050

[30] Nicholas Mayone, Pascal Kunz, Beytullah Yigit, Wissem Soussi, Burkhard Stiller, and Gürkan Gür. 2024. IPv6 Connection Shuffling for Moving Target Defense (MTD) in SDN. In *2024 IEEE International Conference on Cyber Security and Resilience (CSR)*. 373–378. doi:10.1109/CSR61664.2024.10679400

[31] Nick McKeown et al. 2008. OpenFlow: enabling innovation in campus networks. In *ACM SIGCOMM*.

[32] Tina Moghaddam, Guowei Yang, Chandra Thapa, et al. 2024. MTDSense: AI-Based Fingerprinting of Moving Target Defense Techniques in Software-Defined Networking. *arXiv preprint* (2024). arXiv:2408.03758 [cs.CR]

[33] Tina Moghaddam, Guowei Yang, Chandra Thapa, et al. 2024. MTDSense: AI-Based Fingerprinting of Moving Target Defense Techniques in Software-Defined Networking. *arXiv preprint* abs/2408.03758 (2024). https://arxiv.org/abs/2408.03758

[34] J. Narantuya et al. 2019. SDN-Based IP Shuffling Moving Target Defense with Multiple SDN Controllers. In *2019 49th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*. doi:10.1109/DSN-S.2019.00013

[35] Jargalsaikhan Narantuya, Seunghyun Yoon, Hyuk Lim, Jin-Hee Cho, Dong Seong Kim, Terrence Moore, and Frederica Nelson. 2019. SDN-Based IP Shuffling Moving Target Defense with Multiple SDN Controllers. In *2019 49th Annual IEEE/IFIP International Conference on Dependable Systems and Networks – Supplemental Volume (DSN-S)*. 15–16. doi:10.1109/DSN-S.2019.00013

[36] Achintya Prakash and Michael P. Wellman. 2015. Empirical Game-Theoretic Analysis for Moving Target Defense. 9 pages. doi:10.1145/2808475.2808483

[37] Achintya Prakash and Michael P. Wellman. 2015. Empirical Game-Theoretic Analysis for Moving Target Defense. In *Proceedings of the Second ACM Workshop on Moving Target Defense* (Denver, Colorado, USA) *(MTD '15)*. Association for Computing Machinery, New York, NY, USA, 57–65. doi:10.1145/2808475.2808483

[38] Achintya Prakash and Michael P. Wellman. 2015. Empirical Game-Theoretic Analysis for Moving Target Defense. In *Proceedings of the 2nd ACM Workshop on Moving Target Defense*. doi:10.1145/2808475.2808483

[39] Sailik Sengupta, Ankur Chowdhary, Abdulhakim Sabur, Adel Alshamrani, Dijiang Huang, and Subbarao Kambhampati. 2020. A survey of moving target defenses for network security. *IEEE Communications Surveys & Tutorials* 22, 3 (2020), 1909–1941.

[40] Jessica Steinberger, Benjamin Kuhnert, Christian Dietz, Lisa Ball, Anna Sperotto, Harald Baier, Aiko Pras, and Gabi Dreo. 2018. DDoS defense using MTD and SDN. In *NOMS 2018 - 2018 IEEE/IFIP Network Operations and Management Symposium*. 1–9. doi:10.1109/NOMS.2018.8406221

[41] Jinglei Tan, Hui Jin, Hongqi Zhang, Yuchen Zhang, Dexian Chang, Xiaohu Liu, and Hengwei Zhang. 2023. A survey: When moving target defense meets game theory. *Comput. Sci. Rev.* 48, C (May 2023), 19 pages. doi:10.1016/j.cosrev.2023.100544

[42] Hengwei Zhang, Jinglei Tan, Xiaohu Liu, and Jindong Wang. 2020. Moving Target Defense Decision-Making Method: A Dynamic Markov Differential Game Model. In *Proceedings of the 7th ACM Workshop on Moving Target Defense* (Virtual Event, USA) *(MTD'20)*. Association for Computing Machinery, New York, NY, USA, 21–29. doi:10.1145/3411496.3421222

[43] Yuyang Zhou, Guang Cheng, Kang Du, Zihan Chen, Tian Qin, and Yuyu Zhao. 2025. From Static to Adaptive Defense: Federated Multi-Agent Deep Reinforcement Learning-Driven Moving Target Defense. *IEEE Transactions on Cognitive Communications and Networking* (2025). doi:10.1109/TCCN.2025.3630007

[44] Yuyang Zhou, Guang Cheng, Kang Du, Zihan Chen, Tian Qin, and Yuyu Zhao. 2025. From Static to Adaptive Defense: Federated Multi-Agent Deep Reinforcement Learning-Driven Moving Target Defense Against DoS Attacks in UAV Swarm Networks. *IEEE Transactions on Cognitive Communications and Networking* 12 (2025), 3407–3421. doi:10.1109/TCCN.2025.3630007

[45] Ronghai Zhuang, Scott DeLoach, and Xinming Ou. 2014. A survey of moving target defense techniques for cyber systems. *IEEE Communications Surveys & Tutorials* (2014).

A Appendix

A.1 Detailed Network Configuration

A.2 Attack Scripts

A.3 Additional Experimental Results

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